SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

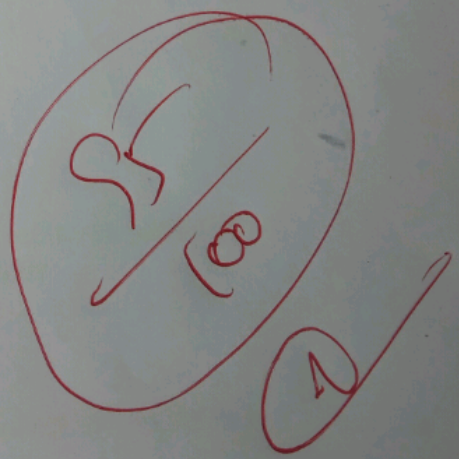
QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Bharath M | 192324302 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RSA Key generation and Security Analysis:

Given Data:

- Prime number $p = 17$
- Prime number $q = 19$
- Public expondent $e = 5$
- Plain text message $M = 7$

1. Calculate n and $\phi(n)$.
2. Calculate the private key d
3. Encrypt the message $M = 7$.
4. Decrypt the ciphertext and verify the result.

Formula: $n = p \times q$

$$n = 17 \times 19$$

$$n = 323$$

The modulus is $n = 323$

Euler's totient function is calculated using the formula,

$$\phi(n) = (p-1)(q-1)$$

Substitute $p = 17$ and $q = 19$:

$$\phi(n) = (17-1)(19-1).$$

$$\phi(n) = 16 \times 18$$

$$\phi(n) = 288$$

Calculate the private key d

Formula: $d \times e \equiv 1 \pmod{\phi(n)}$

Given: $e = 5$, $\phi(n) = 288$

Therefore: $5d \equiv 1 \pmod{288}$

This means we need to find an integer d

$$5d = 1 + 288k$$

Try $k = 1$:

$$5d = 1 + 288$$

$$5d = 289$$

$$d = \frac{289}{5} \times 57.8$$

This is not an integer, so try $k = 2$:

Instead, use the Extended Euclidean Algorithm.

$$5d \equiv 1 \pmod{288}$$

$288 = 5(57) + 3$

$5 = 3(1) + 2$

$3 = 2(1) + 1$

$1 = 3 - 2$

$2 = 5 - 3$

$1 = 3 - (5 - 3)$

$1 = 2(3) - 5$

$-115(5) \equiv 1 \pmod{288}$

$d \equiv -115 \pmod{288}$

$d = 288 - 115$

$\boxed{d = 173}$

$e \times d \equiv 1 \pmod{\phi(n)}$

$5 \times 173 = 865$

RSA keys

$(e,n) = (5, 323)$

The private key is:

$(d,n) = (173, 323)$

Encrypt the Message $M = 7$

$C = M^e \bmod n$

$M = 7$, $e = 5$, $n = 323$

$C = 7^5 \bmod 323$

$7^5 = 16807$

$16807 = 323(52) + 11$

$C = 11$

Decrypt the ciphertext

$C = 11$, $d = 173$, $n = 323$

$M = 11^{173} \bmod 323$

$11^1 \bmod 323 = 11$

$11^2 = 121$

$11^2 \bmod 323 = 121$

$121^2 = 14641$

$14641 = 323(45) + 106$

$11^4 \bmod 323 = 106$

$11^8 \bmod 323 = 254$

power	calculation	Remainder
11^1	11	11
11^2	$11^2 \bmod 323$	121
11^4	$121^2 \bmod 323$	106
11^8	$106^2 \bmod 323$	254
11^{16}	$254^2 \bmod 323$	156
11^{32}	$156^2 \bmod 323$	139
11^{64}	$220^2 \bmod 323$	220

Express the exponent 173 in binary:

$173 = 128 + 32 + 8 + 4 + 1$

$11^{173} = 11^{128} \times 11^{32} \times 11^8 \times 11^4 \times 11$

$M = (121 \times 139 \times 254 \times 106 \times 11) \bmod 323$

Calculate step by step

$121 \times 139 = 16819$

$16819 \bmod 323 = 23$

$23 \times 254 = 5842$

$5842 \bmod 323 = 25$

$25 \times 106 = 2650$

$2650 \bmod 323 = 65$

$65 \times 11 = 715$

$715 \bmod 323 = 69$

$M = 69$

ciphertext was calculated as

$C = 7^5 \bmod 323 = 11$

$11^{173} \bmod 323$

$173 = 128 + 32 + 8 + 4 + 1$

power	Correct remainder
$11^1 \bmod 323$	11
$11^2 \bmod 323$	121
$11^4 \bmod 323$	106
$11^8 \bmod 323$	254
$11^{16} \bmod 323$	156
$11^{32} \bmod 323$	139
$11^{64} \bmod 323$	220
$11^{128} \bmod 323$	121

Select primes

Manual Encryption Analysis

Given Data.

- Prime Number $p = 23$
- Generator $g = 5$
- Private key $x = 7$
- Random number $k = 3$
- plaintext message $M = 10$

Calculate the public key:

(p, g, y)

- p is the prime number
- g is the generator
- x is the private key.
- y is calculated from g and x

$$y = g^x \bmod p$$

$g = 5$, $x = 7$, $p = 23$

$y = 5^7 \bmod 23$

$5^7 = 78125$

$y = 78125 \bmod 23$

$78125 = 23(3396) + 17$

$y = 17$

public:

$(p, g, y) = (23, 5, 17)$

private:

$x = 7$

Encrypt the Message $M = 10$

$$C_1 = g^k \bmod p$$

$$C_2 = M \cdot y^k \bmod p$$

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation